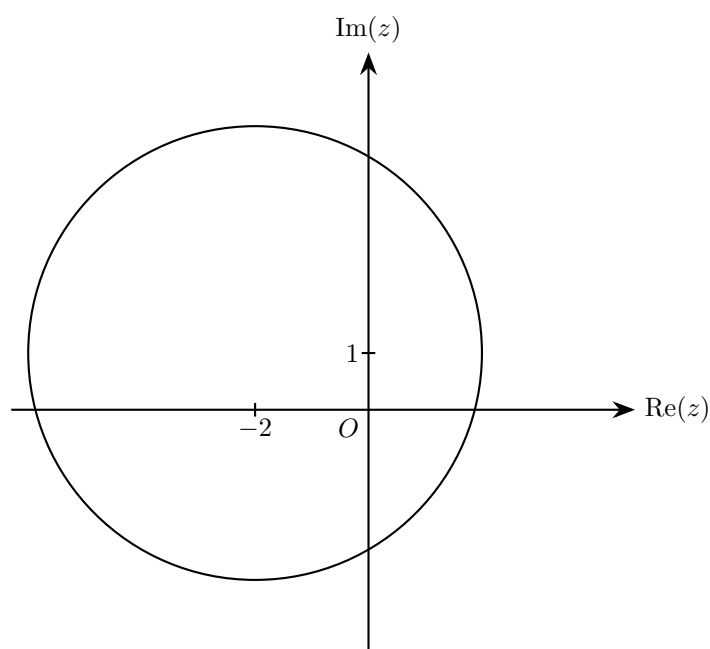


## Questions

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1. The Argand diagram shows a circle of radius 4. The centre of the circle is the point which represents the complex number  $-2 + i$ .

- (a) Use set notation to define the locus of complex numbers,  $z$ , represented by points which lie on the circle. [2]

The locus  $L$  is defined by

$$L = \left\{ z \in \mathbb{C} \mid |z - (2 + i)| = |z + 1 - 3i| \right\}$$

- (b) On a copy of the Argand diagram, sketch and label the locus  $L$ . [2]

You are given that the locus

$$\left\{ z \in \mathbb{C} \mid \arg(z + 1 - 2i) = \frac{2\pi}{3}, \operatorname{Im}(z) = 5 \right\}$$

contains only one number.

- (c) Find this number. [2]

### Solution

- (a) The circle has centre  $-2 + i$  and radius 4.

So the locus is the set of complex numbers whose distance from  $-2 + i$  is 4:

$$\{z \in \mathbb{C} \mid |z - (-2 + i)| = 4\} = \{z \in \mathbb{C} \mid |z + 2 - i| = 4\}$$

- (b) Let  $z = x + iy$ , where  $x, y \in \mathbb{R}$ .

Then

$$|z - (2 + i)| = |z + 1 - 3i|$$

gives

$$|(x - 2) + i(y - 1)| = |(x + 1) + i(y - 3)|$$

Squaring both sides:

$$(x-2)^2 + (y-1)^2 = (x+1)^2 + (y-3)^2$$

Now expand:

$$\begin{aligned} x^2 - 4x + 4 + y^2 - 2y + 1 &= x^2 + 2x + 1 + y^2 - 6y + 9 \\ -4x - 2y + 5 &= 2x - 6y + 10 \\ -6x + 4y - 5 &= 0 \\ 6x - 4y + 5 &= 0 \end{aligned}$$

So

$$L : 6x - 4y + 5 = 0$$

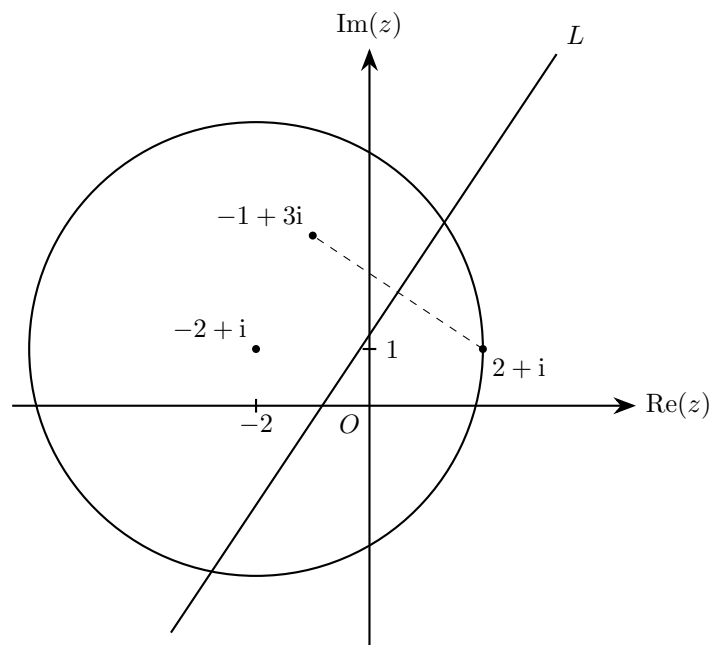
or equivalently

$$y = \frac{3}{2}x + \frac{5}{4}$$

This is a straight line. Since it is the set of points equidistant from  $2 + i$  and  $-1 + 3i$ , it is the perpendicular bisector of the line segment joining those two points.

It passes through the midpoint of that segment,

$$\left( \frac{2 + (-1)}{2}, \frac{1 + 3}{2} \right) = \left( \frac{1}{2}, 2 \right)$$



(c) Since  $\text{Im}(z) = 5$ , write

$$z = x + 5i$$

Then

$$z + 1 - 2i = (x + 1) + 3i$$

We are given

$$\arg(z + 1 - 2i) = \frac{2\pi}{3}$$

So  $(x + 1) + 3i$  lies in the second quadrant, with  $x + 1 < 0$ . Therefore

$$\tan \frac{2\pi}{3} = \frac{3}{x + 1}$$

Using  $\tan \frac{2\pi}{3} = -\sqrt{3}$ ,

$$\frac{3}{x+1} = -\sqrt{3}$$

Hence

$$x+1 = -\sqrt{3}$$

so

$$x = -1 - \sqrt{3}$$

Therefore the required number is

$$z = -1 - \sqrt{3} + 5i$$

2. In an Argand diagram, the point  $P$  representing the complex number  $w$  lies on the locus defined by

$$\left\{ z \in \mathbb{C} \mid \arg(z - 2 + i) = \frac{\pi}{6} \right\}$$

You are given that  $\text{Im}(w) = 3$ .

(a) Find  $w$ .

[2]

The point  $P$  also lies on the locus defined by  $\left\{ z \in \mathbb{C} \mid |z - 2 - 11i| = k \right\}$ , where  $k$  is a constant.

(b) Find the complex number represented by the other point of intersection of the loci defined by

$$\left\{ z \in \mathbb{C} \mid |z - 2 - 11i| = k \right\} \quad \text{and} \quad \left\{ z \in \mathbb{C} \mid \arg(z - 2 + i) = \frac{\pi}{6} \right\}$$

[7]

### Solution

(a) Let

$$w = x + 3i$$

since  $\text{Im}(w) = 3$ .

Then

$$w - 2 + i = (x - 2) + 4i$$

Given

$$\arg(w - 2 + i) = \frac{\pi}{6}$$

so

$$\tan \frac{\pi}{6} = \frac{4}{x - 2}$$

Hence

$$\frac{1}{\sqrt{3}} = \frac{4}{x - 2}$$

so

$$x - 2 = 4\sqrt{3}$$

and therefore

$$x = 2 + 4\sqrt{3}$$

Thus

$$w = 2 + 4\sqrt{3} + 3i$$

(b) From part (a),

$$P \text{ represents } w = 2 + 4\sqrt{3} + 3i$$

The second locus is

$$|z - 2 - 11i| = k = |z - (2 + 11i)|$$

which is a circle with centre  $2 + 11i$ .

Using the point  $P$  to find  $k$ ,

$$\begin{aligned} k &= |w - (2 + 11i)| \\ &= |(2 + 4\sqrt{3} + 3i) - (2 + 11i)| \\ &= |4\sqrt{3} - 8i| \end{aligned}$$

So

$$\begin{aligned}k^2 &= (4\sqrt{3})^2 + (-8)^2 \\&= 48 + 64 \\&= 112\end{aligned}$$

and hence

$$k = 4\sqrt{7}$$

Now let a general point of intersection be

$$z = x + yi$$

Since

$$\arg(z - 2 + i) = \frac{\pi}{6}$$

the point lies on the half-line starting at  $(2, -1)$  with gradient

$$\tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$$

so its equation is

$$y + 1 = \frac{x - 2}{\sqrt{3}}$$

that is,

$$y = \frac{x - 2}{\sqrt{3}} - 1$$

The circle has equation

$$(x - 2)^2 + (y - 11)^2 = 112$$

Substitute  $y = \frac{x - 2}{\sqrt{3}} - 1$ :

$$(x - 2)^2 + \left(\frac{x - 2}{\sqrt{3}} - 12\right)^2 = 112$$

Let

$$u = x - 2$$

Then

$$u^2 + \left(\frac{u}{\sqrt{3}} - 12\right)^2 = 112$$

Expanding,

$$\begin{aligned}u^2 + \left(\frac{u^2}{3} - 8\sqrt{3}u + 144\right) &= 112 \\ \frac{4u^2}{3} - 8\sqrt{3}u + 144 &= 112 \\ \frac{4u^2}{3} - 8\sqrt{3}u + 32 &= 0\end{aligned}$$

Multiply by 3:

$$4u^2 - 24\sqrt{3}u + 96 = 0$$

Divide by 4:

$$u^2 - 6\sqrt{3}u + 24 = 0$$

Factorising,

$$(u - 4\sqrt{3})(u - 2\sqrt{3}) = 0$$

So

$$u = 4\sqrt{3} \quad \text{or} \quad u = 2\sqrt{3}$$

Both values are positive, so both points have  $x > 2$  and lie on the half-line itself, not on its extension.

The value  $u = 4\sqrt{3}$  gives the known point  $P$  from part (a), so the other intersection has

$$u = 2\sqrt{3}$$

Therefore

$$x = 2 + 2\sqrt{3}$$

and

$$y = \frac{2\sqrt{3}}{\sqrt{3}} - 1 = 2 - 1 = 1$$

Hence the other point represents

$$z = 2 + 2\sqrt{3} + \mathrm{i}$$

3. On an Argand diagram, the locus of points satisfying  $|z - 4 + 3i| = 8$  is a circle.

(a) Write down

(i) the coordinates of the centre of this circle

[1]

(ii) the radius of this circle

[1]

(b) On an Argand diagram, shade the set of points defined by

$$|z - 4 + 3i| \leq 8$$

[1]

(c) For the set of points defined in part (b), determine the maximum value of  $|z|$ .

[3]

The set of points  $A$  is defined by

$$A = \left\{ z \in \mathbb{C} \mid -\frac{\pi}{4} \leq \arg(z - 3 - 6i) \leq \frac{3\pi}{4} \right\} \cap \left\{ z \in \mathbb{C} \mid |z - 4 + 3i| \leq 8 \right\}$$

(d) Determine the area of the region defined by  $A$ , giving your answer to 3 significant figures.

[4]

### Solution

(a) (i)

$$|z - 4 + 3i| = |z - (4 - 3i)|$$

So the circle has centre  $(4, -3)$ .

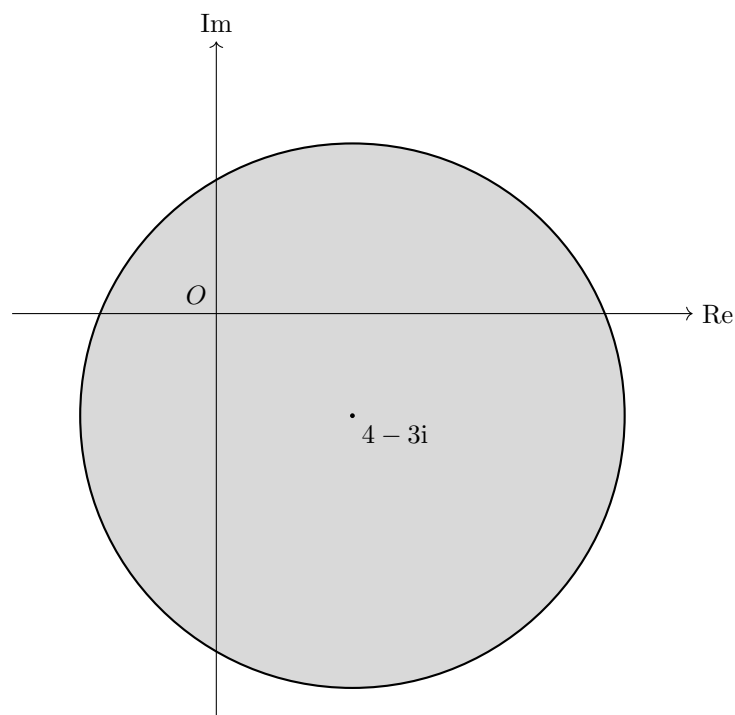
(ii) Comparing with the standard form  $|z - a| = r$ , the radius is 8.

(b) The inequality

$$|z - 4 + 3i| \leq 8$$

means all points whose distance from  $4 - 3i$  is at most 8.

So we shade the interior and boundary of the circle with centre  $(4, -3)$  and radius 8.



- (c) The modulus  $|z|$  is the distance from the origin  $O$  to the point representing  $z$ .

The centre  $C$  of the circle represents  $4 - 3i$ , so

$$OC = |4 - 3i| = \sqrt{4^2 + (-3)^2} = 5$$

The point of the region furthest from  $O$  is where the line from  $O$  through  $C$  meets the circle on the far side of  $C$ . Its distance from  $O$  is  $OC$  plus the radius.

Therefore the maximum value of  $|z|$  is

$$5 + 8 = 13$$

- (d) Let  $z = x + iy$ .

First consider the argument condition:

$$-\frac{\pi}{4} \leq \arg(z - 3 - 6i) \leq \frac{3\pi}{4}$$

Since

$$z - 3 - 6i = (x - 3) + i(y - 6)$$

the boundary arguments  $-\pi/4$  and  $3\pi/4$  are opposite directions, so they form a straight line through  $(3, 6)$  with gradient  $-1$ .

Hence the boundary line is

$$y - 6 = -(x - 3)$$

so

$$x + y = 9$$

To decide which side is needed, take the point  $(3, 7)$ . Then

$$\arg((3 + 7i) - 3 - 6i) = \arg(i) = \frac{\pi}{2}$$

which is in the given range, and  $(3, 7)$  satisfies  $x + y = 10$ .

So the argument condition is the half-plane

$$x + y \geq 9$$

The circle has centre  $C = (4, -3)$  and radius 8. Since

$$4 + (-3) = 1 < 9$$

the centre lies on the opposite side of the line, so the region  $A$  is the minor segment cut off by the line  $x + y = 9$ .

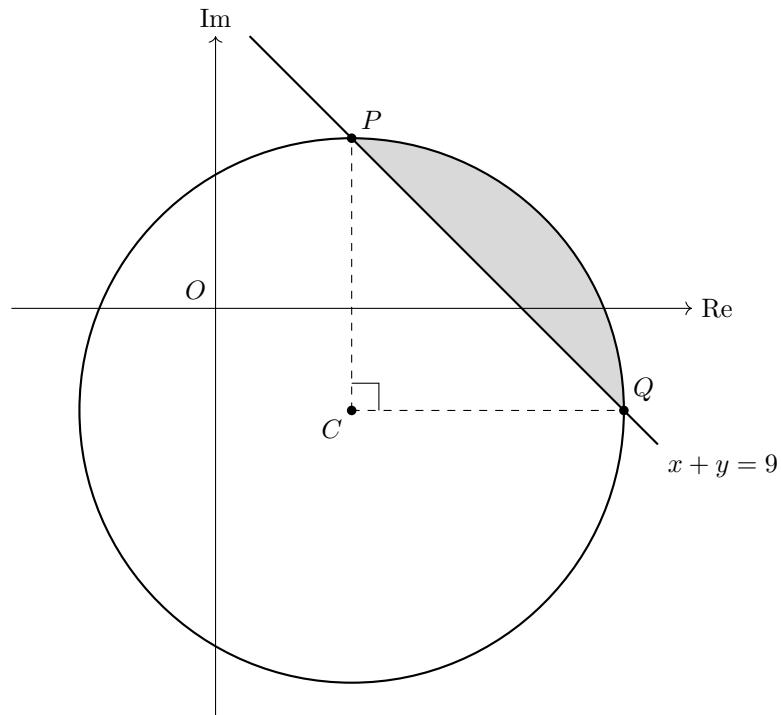
The circle has equation  $(x - 4)^2 + (y + 3)^2 = 64$ . To find where the line meets it, substitute  $y = 9 - x$ :

$$\begin{aligned}(x - 4)^2 + (12 - x)^2 &= 64 \\ x^2 - 8x + 16 + x^2 - 24x + 144 &= 64 \\ 2x^2 - 32x + 96 &= 0 \\ x^2 - 16x + 48 &= 0 \\ (x - 4)(x - 12) &= 0\end{aligned}$$

So the line meets the circle at  $P = (4, 5)$  and  $Q = (12, -3)$ .

$CP$  joins  $(4, -3)$  to  $(4, 5)$ , so it is vertical, and  $CQ$  joins  $(4, -3)$  to  $(12, -3)$ , so it is horizontal. Hence

$$\angle PCQ = \frac{\pi}{2}$$



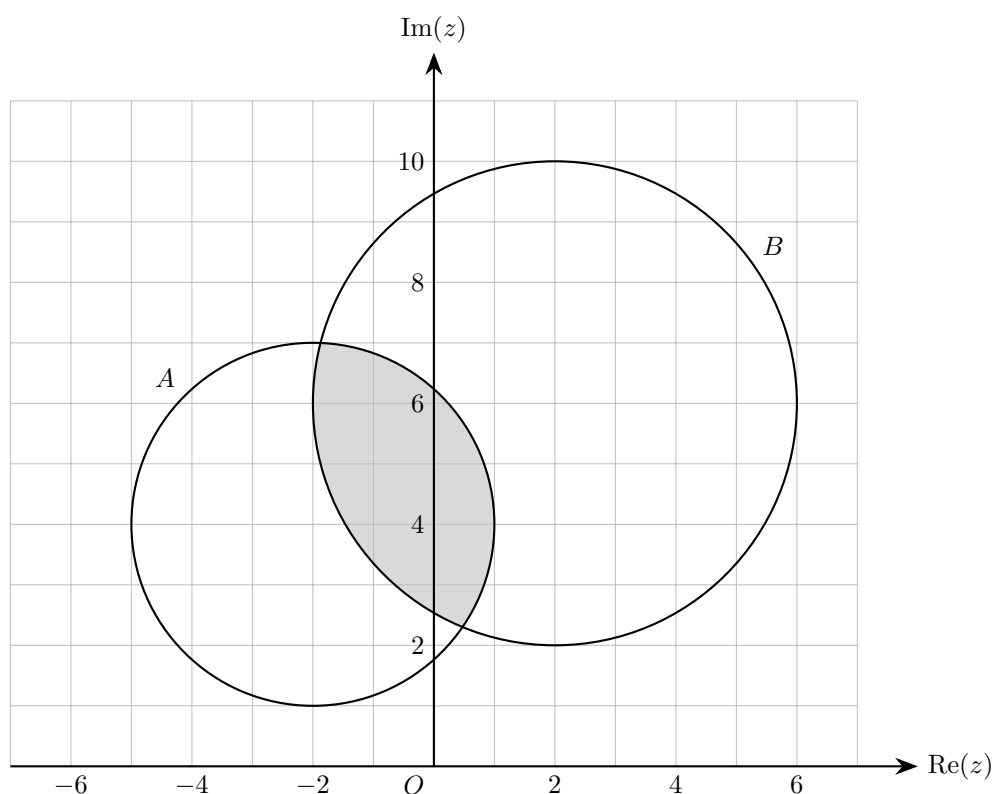
Therefore

$$\begin{aligned}
 \text{Area}(A) &= \text{area of sector } PCQ - \text{area of } \triangle PCQ \\
 &= \frac{1}{2}r^2\theta - \frac{1}{2}r^2 \sin \theta \\
 &= \frac{1}{2} \cdot 8^2 \cdot \frac{\pi}{2} - \frac{1}{2} \cdot 8^2 \cdot \sin \frac{\pi}{2} \\
 &= 16\pi - 32
 \end{aligned}$$

So the area is

$$16\pi - 32 \approx 18.3$$

to 3 significant figures.



4. (a) The diagram above shows an Argand diagram.

(i) Write down the equation of the locus represented by the circumference of circle  $A$ . [3]

(ii) Write down the two inequalities that define the shaded region, not including either circumference. [3]

(b) (i) Draw an Argand diagram to show the region where

$$-\frac{\pi}{6} < \arg(z + 2 - i) < \frac{\pi}{4} \quad [3]$$

(ii) Determine whether the point  $1 + 5i$  lies within this region. [3]

### Solution

(a) (i) For a circle with centre  $z_0$  and radius  $r$ , the locus is

$$|z - z_0| = r$$

Circle  $A$  has centre  $-2 + 4i$  and radius 3, so

$$|z - (-2 + 4i)| = 3$$

hence the circumference of circle  $A$  is

$$|z + 2 - 4i| = 3$$

(ii) The shaded region is the part inside both circles, and neither circumference is included.

Inside circle  $A$ :

$$|z + 2 - 4i| < 3$$

Circle  $B$  has centre  $2 + 6i$  and radius 4, so inside circle  $B$ :

$$|z - (2 + 6i)| < 4$$

So the two inequalities are

$$|z + 2 - 4i| < 3 \quad \text{and} \quad |z - (2 + 6i)| < 4$$

(b) (i) We have

$$z + 2 - i = z - (-2 + i)$$

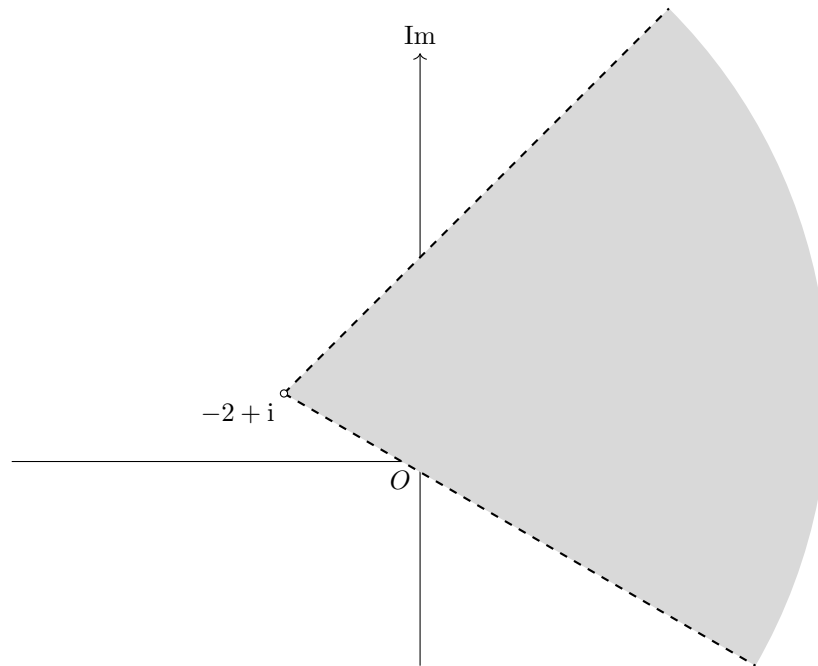
so the argument is measured from the point  $-2 + i$ .

The region

$$-\frac{\pi}{6} < \arg(z + 2 - i) < \frac{\pi}{4}$$

is therefore the open sector with vertex at  $-2 + i$  between the rays making angles  $-\frac{\pi}{6}$  and  $\frac{\pi}{4}$  with the positive real axis.

Since the inequalities are strict, both boundary rays are excluded. The vertex  $-2 + i$  is excluded too, as  $\arg 0$  is undefined.



(ii) For the point  $z = 1 + 5i$ ,

$$z + 2 - i = (1 + 5i) + 2 - i = 3 + 4i$$

So

$$\arg(z + 2 - i) = \arg(3 + 4i) = \tan^{-1}\left(\frac{4}{3}\right) \approx 0.927$$

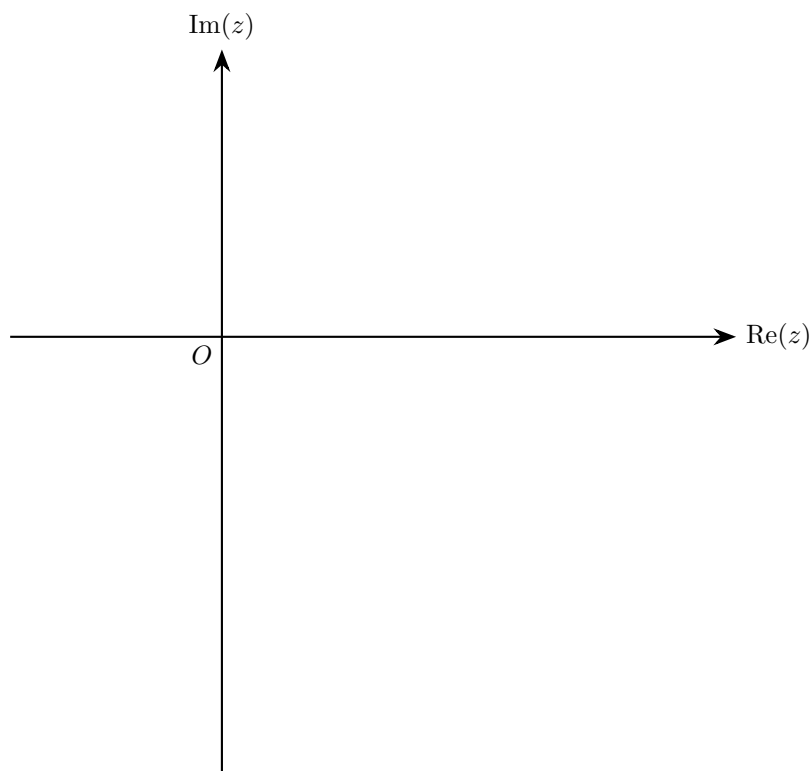
But

$$\frac{\pi}{4} \approx 0.785$$

Since

$$0.927 > \frac{\pi}{4}$$

the argument is too large, so  $1 + 5i$  does not lie in the region.



5. (a) Find the modulus of the complex number  $2 + 2\sqrt{3}i$ , giving your answer as an integer. [2]
- (b) The locus of points,  $L$ , satisfies the equation  $|z - (2 - 2\sqrt{3}i)| = 2$ .  
 (i) Sketch the locus  $L$  on the Argand diagram below. [3]  
 (ii) The complex number  $w$  lies on  $L$  so that  $-\pi < \arg w \leq \pi$ . Find the greatest possible value of  $\arg w$ , giving your answer in terms of  $\pi$ . [2]
- (c) Solve the equation  $z^4 = 2 + 2\sqrt{3}i$ , giving your answers in the form  $re^{i\theta}$ , where  $r > 0$  and  $-\pi < \theta \leq \pi$ . [5]

### Solution

- (a) The modulus is

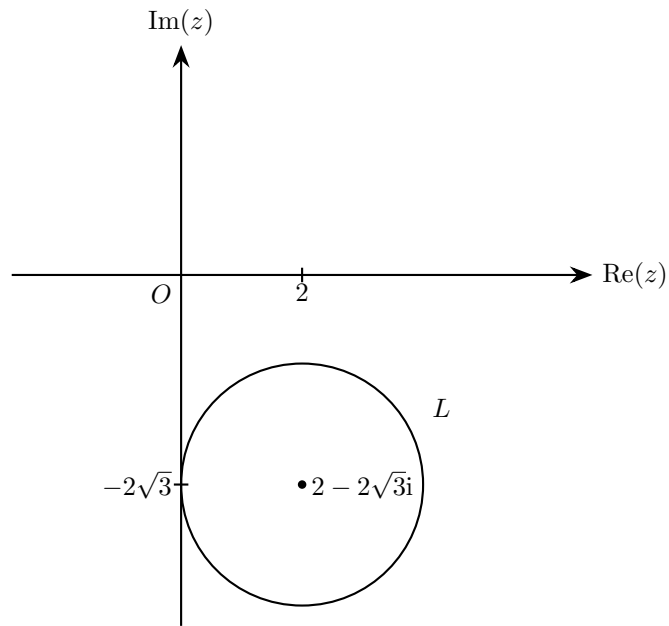
$$|2 + 2\sqrt{3}i| = \sqrt{2^2 + (2\sqrt{3})^2} = \sqrt{4 + 12} = \sqrt{16} = 4$$

- (b) (i) The equation

$$|z - (2 - 2\sqrt{3}i)| = 2$$

represents all points a distance 2 from the point  $2 - 2\sqrt{3}i$ .

So  $L$  is a circle with centre  $(2, -2\sqrt{3})$  and radius 2. The centre is a distance 2 from the imaginary axis, so the circle touches the imaginary axis at  $-2\sqrt{3}i$ .



(ii) Let  $C$  be the centre of the circle, so

$$C = 2 - 2\sqrt{3}i$$

First find  $OC$ :

$$OC = |2 - 2\sqrt{3}i| = \sqrt{2^2 + (-2\sqrt{3})^2} = \sqrt{4 + 12} = 4$$

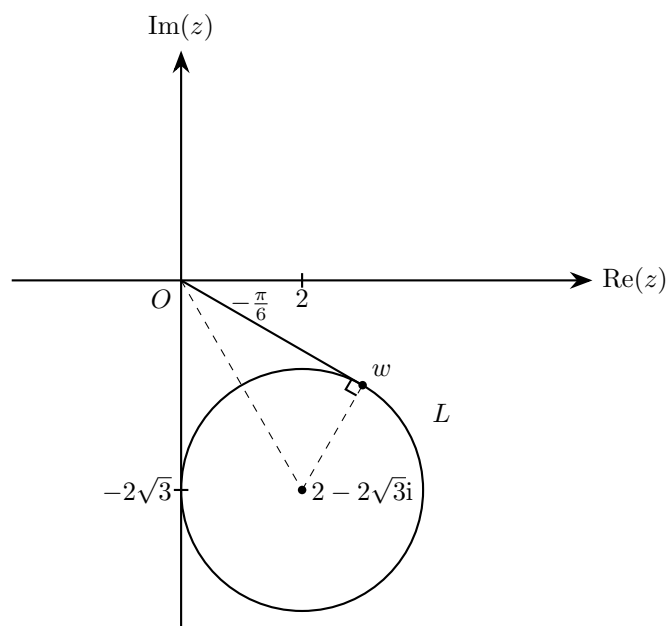
Also,

$$\arg C = -\frac{\pi}{3}$$

because  $C$  is in the fourth quadrant and

$$\tan |\arg C| = \frac{2\sqrt{3}}{2} = \sqrt{3}$$

The greatest possible value of  $\arg w$  occurs when the line from the origin to the circle is a tangent, touching the circle at  $w$ .



Then triangle  $OCW$  is right-angled at  $W$ , with

$$CW = 2, \quad OC = 4$$

Let  $\angle COW = \alpha$ . Then

$$\sin \alpha = \frac{CW}{OC} = \frac{2}{4} = \frac{1}{2}$$

so

$$\alpha = \frac{\pi}{6}$$

Since this tangent is above the line  $OC$ ,

$$\arg w = \arg C + \alpha = -\frac{\pi}{3} + \frac{\pi}{6} = -\frac{\pi}{6}$$

Therefore the greatest possible value of  $\arg w$  is  $-\frac{\pi}{6}$ .

(c) From part (a),

$$|2 + 2\sqrt{3}i| = 4$$

and since  $2 + 2\sqrt{3}i$  lies in the first quadrant,

$$\arg(2 + 2\sqrt{3}i) = \frac{\pi}{3}$$

So

$$2 + 2\sqrt{3}i = 4e^{i\pi/3}$$

We solve

$$z^4 = 4e^{i\pi/3}$$

Let  $z = re^{i\theta}$ . Then

$$z^4 = r^4 e^{i4\theta}$$

so

$$r^4 e^{i4\theta} = 4e^{i(\pi/3+2k\pi)} \quad (k \in \mathbb{Z})$$

Equating moduli gives

$$r^4 = 4$$

so, with  $r > 0$ ,

$$r = 4^{1/4} = \sqrt{2}$$

Equating arguments gives

$$4\theta = \frac{\pi}{3} + 2k\pi$$

hence

$$\theta = \frac{\pi}{12} + \frac{k\pi}{2}$$

The four distinct roots come from  $k = 0, 1, 2, 3$ :

$$\theta = \frac{\pi}{12}, \quad \frac{7\pi}{12}, \quad \frac{13\pi}{12}, \quad \frac{19\pi}{12}$$

Writing these in the range  $-\pi < \theta \leq \pi$ ,

$$\theta = \frac{\pi}{12}, \quad \frac{7\pi}{12}, \quad -\frac{11\pi}{12}, \quad -\frac{5\pi}{12}$$

Therefore

$$z = \sqrt{2}e^{i\pi/12}, \quad \sqrt{2}e^{i7\pi/12}, \quad \sqrt{2}e^{-i11\pi/12}, \quad \sqrt{2}e^{-i5\pi/12}$$

6. The point  $P$  represents a complex number  $z$  on an Argand diagram such that

$$|z - (1 + i)| = \frac{1}{2}|z - (4 + i)|$$

- (a) Show that, as  $z$  varies, the locus of  $P$  is a circle, and give the coordinates of the centre and the radius of the circle. [5]

The point  $Q$  represents a complex number  $z$  on an Argand diagram such that

$$|z - (2 + 3i)| = |z + 2 - i|$$

- (b) Sketch, on the same Argand diagram, the locus of  $P$  and the locus of  $Q$  as  $z$  varies. [5]

- (c) On your diagram shade the region which satisfies

$$|z - (1 + i)| \leq \frac{1}{2}|z - (4 + i)| \quad \text{and} \quad |z - (2 + 3i)| \geq |z + 2 - i| \quad [2]$$

### Solution

- (a) Let

$$z = x + iy$$

Then

$$|z - (1 + i)| = \sqrt{(x - 1)^2 + (y - 1)^2}$$

and

$$|z - (4 + i)| = \sqrt{(x - 4)^2 + (y - 1)^2}$$

So the given condition becomes

$$\sqrt{(x - 1)^2 + (y - 1)^2} = \frac{1}{2}\sqrt{(x - 4)^2 + (y - 1)^2}$$

Squaring both sides,

$$4((x - 1)^2 + (y - 1)^2) = (x - 4)^2 + (y - 1)^2$$

Expand:

$$\begin{aligned} 4(x^2 - 2x + 1 + y^2 - 2y + 1) &= x^2 - 8x + 16 + y^2 - 2y + 1 \\ 4x^2 - 8x + 4 + 4y^2 - 8y + 4 &= x^2 - 8x + 16 + y^2 - 2y + 1 \end{aligned}$$

Bring everything to one side:

$$\begin{aligned} 3x^2 + 3y^2 - 6y - 9 &= 0 \\ x^2 + y^2 - 2y - 3 &= 0 \end{aligned}$$

Complete the square in  $y$ :

$$\begin{aligned} x^2 + (y^2 - 2y + 1) - 4 &= 0 \\ x^2 + (y - 1)^2 &= 4 \end{aligned}$$

This is the equation of a circle.

Hence the locus is the circle with centre  $(0, 1)$  and radius 2.

(b) For the second locus, let again  $z = x + iy$ .

Then

$$|z - (2 + 3i)| = \sqrt{(x - 2)^2 + (y - 3)^2}$$

and

$$|z + 2 - i| = |z - (-2 + i)| = \sqrt{(x + 2)^2 + (y - 1)^2}$$

So

$$\sqrt{(x - 2)^2 + (y - 3)^2} = \sqrt{(x + 2)^2 + (y - 1)^2}$$

Squaring both sides,

$$(x - 2)^2 + (y - 3)^2 = (x + 2)^2 + (y - 1)^2$$

Expand and simplify:

$$\begin{aligned}x^2 - 4x + 4 + y^2 - 6y + 9 &= x^2 + 4x + 4 + y^2 - 2y + 1 \\-4x - 6y + 13 &= 4x - 2y + 5 \\-8x - 4y + 8 &= 0 \\2x + y - 2 &= 0\end{aligned}$$

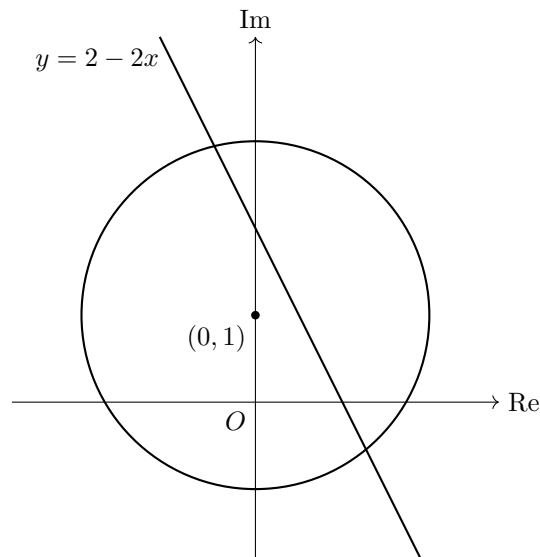
So the locus of  $Q$  is the straight line

$$y = 2 - 2x$$

This is the perpendicular bisector of the line segment joining  $(2, 3)$  and  $(-2, 1)$ .

On the same Argand diagram, we sketch:

- the circle  $x^2 + (y - 1)^2 = 4$ , centre  $(0, 1)$ , radius 2,
- the line  $y = 2 - 2x$ .



(c) From part (a),

$$|z - (1 + i)| \leq \frac{1}{2}|z - (4 + i)|$$

gives

$$x^2 + (y - 1)^2 \leq 4$$

so the point lies inside or on the circle.

From part (b),

$$|z - (2 + 3i)| \geq |z + 2 - i|$$

gives

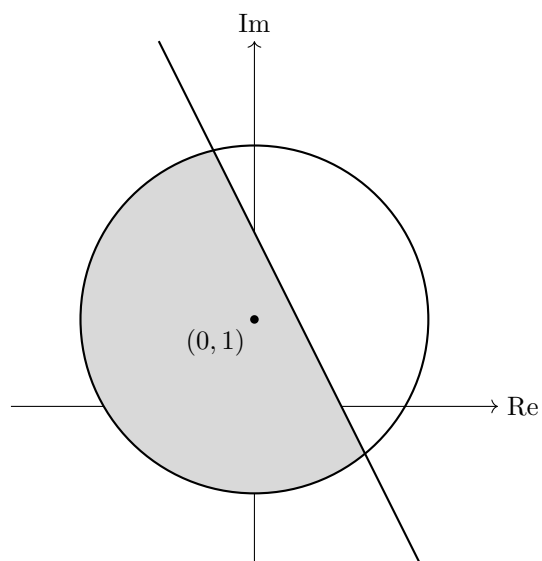
$$\begin{aligned}(x-2)^2 + (y-3)^2 &\geq (x+2)^2 + (y-1)^2 \\ -8x - 4y + 8 &\geq 0 \\ 2x + y - 2 &\leq 0 \\ y &\leq 2 - 2x\end{aligned}$$

So the point lies on or below the line  $y = 2 - 2x$ .

Therefore the required region is the intersection of:

- the inside of the circle  $x^2 + (y-1)^2 = 4$ ,
- the half-plane  $y \leq 2 - 2x$ .

The shaded region is shown below.



7. On an Argand diagram, the point  $P$  representing the complex number  $z$  lies on the locus

$$|z - (2 - i)| = 3$$

The point  $Q$  representing the complex number  $w$  lies on the locus

$$|w + 1 - 11i| = 3|w + 1 - 3i|$$

Find the exact coordinates of the points common to these two loci.

[7]

### Solution

Let a point common to both loci represent the complex number

$$u = x + yi$$

so its coordinates are  $(x, y)$ . (A new letter is used because the question uses  $z$  and  $w$  for the two loci.)  
For the locus of  $P$ ,

$$|u - (2 - i)| = 3$$

Now

$$u - (2 - i) = (x - 2) + (y + 1)i$$

so

$$|(x - 2) + (y + 1)i| = 3$$

Using  $|a + bi| = \sqrt{a^2 + b^2}$ ,

$$(x - 2)^2 + (y + 1)^2 = 9$$

For the locus of  $Q$ ,

$$|u + 1 - 11i| = 3|u + 1 - 3i|$$

Now

$$u + 1 - 11i = (x + 1) + (y - 11)i, \quad u + 1 - 3i = (x + 1) + (y - 3)i$$

So

$$|(x + 1) + (y - 11)i| = 3|(x + 1) + (y - 3)i|$$

Squaring both sides gives

$$(x + 1)^2 + (y - 11)^2 = 9((x + 1)^2 + (y - 3)^2)$$

Expand and simplify:

$$x^2 + 2x + 1 + y^2 - 22y + 121 = 9x^2 + 18x + 9 + 9y^2 - 54y + 81$$

$$0 = 8x^2 + 16x + 8 + 8y^2 - 32y - 40$$

$$0 = x^2 + 2x + y^2 - 4y - 4$$

$$0 = (x + 1)^2 + (y - 2)^2 - 9$$

Hence the second locus is

$$(x + 1)^2 + (y - 2)^2 = 9$$

So the common points satisfy the simultaneous equations

$$(x - 2)^2 + (y + 1)^2 = 9$$

and

$$(x + 1)^2 + (y - 2)^2 = 9$$

Both right-hand sides equal 9, so equating the left-hand sides:

$$(x - 2)^2 + (y + 1)^2 = (x + 1)^2 + (y - 2)^2$$

$$x^2 - 4x + 4 + y^2 + 2y + 1 = x^2 + 2x + 1 + y^2 - 4y + 4$$

$$-6x + 6y = 0$$

$$y = x$$

Substitute  $y = x$  into  $(x - 2)^2 + (y + 1)^2 = 9$ :

$$\begin{aligned}(x - 2)^2 + (x + 1)^2 &= 9 \\ x^2 - 4x + 4 + x^2 + 2x + 1 &= 9 \\ 2x^2 - 2x - 4 &= 0 \\ x^2 - x - 2 &= 0 \\ (x - 2)(x + 1) &= 0\end{aligned}$$

So

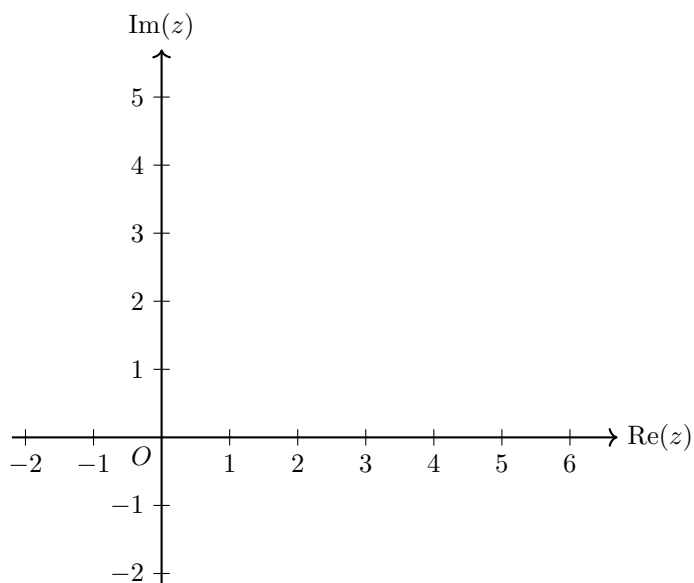
$$x = 2 \quad \text{or} \quad x = -1$$

and since  $y = x$ ,

$$y = 2 \quad \text{or} \quad y = -1$$

Therefore the exact coordinates of the common points are

$$(2, 2) \quad \text{and} \quad (-1, -1)$$



8. (a) Sketch, on the Argand diagram above, the locus of points satisfying the equation

$$|z - (3 + i)| = 1 \quad [1]$$

- (b) There is a unique complex number  $w$  that satisfies both

$$|w - (3 + i)| = 1 \quad \text{and} \quad \arg(w) = \alpha$$

where  $\alpha$  is a constant such that  $0 < \alpha < \pi$ .

- (i) Find the value of  $\alpha$ . [2]  
 (ii) Find  $w$  and express it in the form  $r(\cos \theta + i \sin \theta)$ . Give each of  $r$  and  $\theta$  to two significant figures. [4]

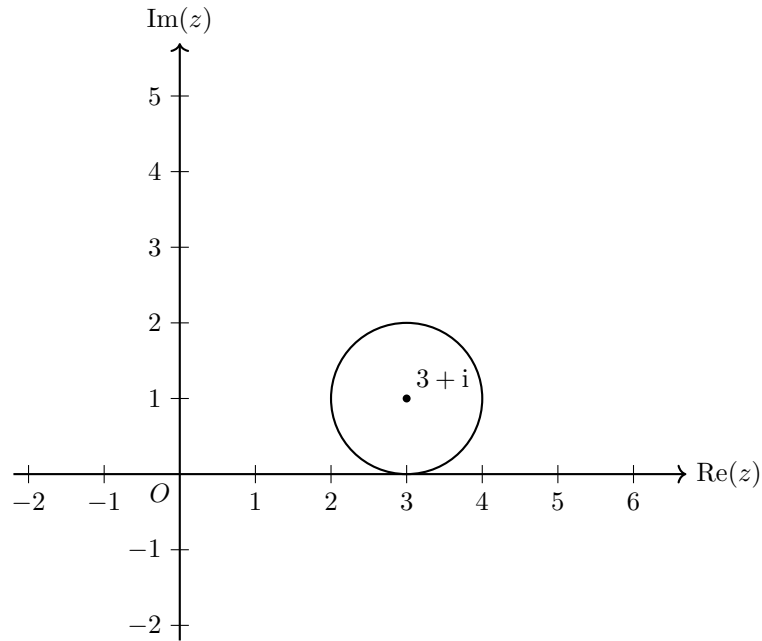
### Solution

- (a) The locus

$$|z - (3 + i)| = 1$$

is the set of points whose distance from  $3 + i$  is 1.

So it is a circle with centre  $(3, 1)$  and radius 1.



- (b) (i) If  $\arg(w) = \alpha$ , then  $w$  lies on the half-line from the origin at angle  $\alpha$ . This half-line is part of the line

$$y = mx$$

where

$$m = \tan \alpha$$

For there to be a unique point  $w$  satisfying both conditions, the half-line must touch the circle at exactly one point, so it must be a tangent.

The circle has equation  $(x - 3)^2 + (y - 1)^2 = 1$ . Substituting  $y = mx$ ,

$$\begin{aligned}(x - 3)^2 + (mx - 1)^2 &= 1 \\ x^2 - 6x + 9 + m^2x^2 - 2mx + 1 &= 1 \\ (1 + m^2)x^2 - (6 + 2m)x + 9 &= 0\end{aligned}$$

For a tangent this quadratic has a repeated root, so its discriminant is zero:

$$\begin{aligned}(6 + 2m)^2 - 4(1 + m^2)(9) &= 0 \\ 36 + 24m + 4m^2 - 36 - 36m^2 &= 0 \\ 24m - 32m^2 &= 0 \\ 8m(3 - 4m) &= 0\end{aligned}$$

So

$$m = 0 \quad \text{or} \quad m = \frac{3}{4}$$

If  $m = 0$ , then  $\alpha = 0$  (the real axis, which touches the circle at 3), but the question states  $0 < \alpha < \pi$ , so this is not allowed.

Hence

$$\tan \alpha = \frac{3}{4}$$

so

$$\alpha = \arctan\left(\frac{3}{4}\right) \approx 0.644 \text{ rad}$$

- (ii) Using the value of  $m$ , the tangent line is

$$y = \frac{3}{4}x$$

The circle equation is

$$(x - 3)^2 + (y - 1)^2 = 1$$

Substituting  $y = \frac{3}{4}x$ ,

$$\begin{aligned}(x - 3)^2 + \left(\frac{3}{4}x - 1\right)^2 &= 1 \\ x^2 - 6x + 9 + \frac{9}{16}x^2 - \frac{3}{2}x + 1 &= 1 \\ \frac{25}{16}x^2 - \frac{15}{2}x + 9 &= 0\end{aligned}$$

Multiplying by 16,

$$25x^2 - 120x + 144 = 0$$

Factorising,

$$(5x - 12)^2 = 0$$

So

$$x = \frac{12}{5}$$

Then

$$y = \frac{3}{4} \cdot \frac{12}{5} = \frac{9}{5}$$

Hence

$$w = \frac{12}{5} + \frac{9}{5}\mathbf{i}$$

Now find its modulus:

$$\begin{aligned}|w| &= \sqrt{\left(\frac{12}{5}\right)^2 + \left(\frac{9}{5}\right)^2} \\ &= \sqrt{\frac{144}{25} + \frac{81}{25}} \\ &= \sqrt{\frac{225}{25}} \\ &= 3\end{aligned}$$

Its argument is

$$\theta = \arg(w) = \alpha = \arctan\left(\frac{3}{4}\right) \approx 0.64$$

So, with  $r = 3.0$  and  $\theta = 0.64$  to two significant figures,

$$w = \frac{12}{5} + \frac{9}{5}\mathbf{i} = 3.0(\cos 0.64 + \mathbf{i} \sin 0.64)$$

9. The locus of points  $z = x + iy$  that satisfy

$$\arg\left(\frac{z - (2 + i)}{z - (6 + 5i)}\right) = \frac{\pi}{4}$$

is an arc of a circle  $C$ .

- (a) On an Argand diagram sketch the locus of  $z$ . [2]
- (b) Explain why the centre of  $C$  lies on the line  $y = 7 - x$ . [1]
- (c) Determine the radius of  $C$ . [2]
- (d) Determine the coordinates of the centre of  $C$ . [2]

### Solution

(a) Let

$$A = 2 + i \quad \text{and} \quad B = 6 + 5i$$

so  $A = (2, 1)$  and  $B = (6, 5)$ .

The condition

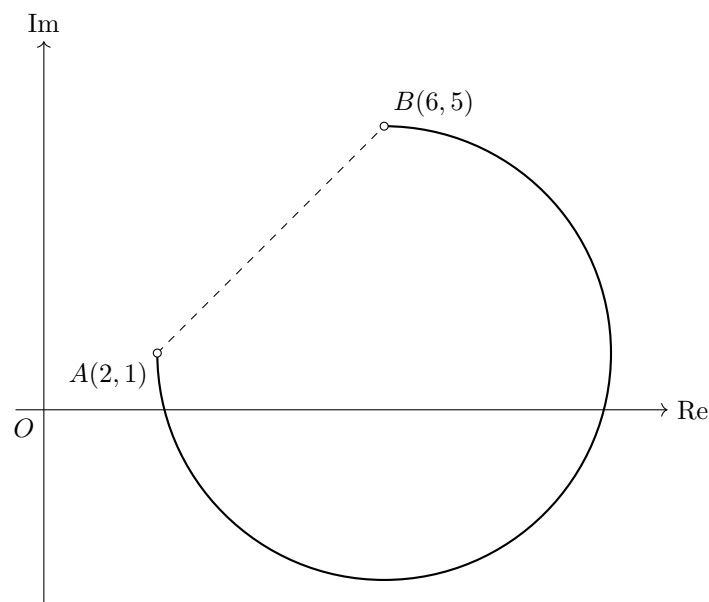
$$\arg\left(\frac{z - A}{z - B}\right) = \frac{\pi}{4}$$

means

$$\arg(z - A) - \arg(z - B) = \frac{\pi}{4}$$

so, if  $Z$  is the point representing  $z$ , the angle  $AZB$  is  $\frac{\pi}{4}$ .

Angles in the same segment are equal, so the locus is an arc of a circle through  $A$  and  $B$ . Since  $\frac{\pi}{4} < \frac{\pi}{2}$ , it is the major arc. It lies on the side below/right of the line segment  $AB$ , as checked in part (d). Also,  $A$  and  $B$  are excluded because the expression is undefined there.



- (b) Since  $A$  and  $B$  lie on circle  $C$ , the centre of  $C$  is equidistant from  $A$  and  $B$ . Therefore it lies on the perpendicular bisector of  $AB$ .

The midpoint of  $AB$  is

$$\left(\frac{2+6}{2}, \frac{1+5}{2}\right) = (4, 3)$$

The gradient of  $AB$  is

$$\frac{5-1}{6-2} = 1$$

so the perpendicular bisector has gradient  $-1$ .

Its equation is

$$y - 3 = -1(x - 4)$$

$$y = 7 - x$$

Therefore the centre lies on the line  $y = 7 - x$ .

(c) Let  $P$  be the centre of  $C$  and  $r$  its radius.

The angle at the circumference is  $\frac{\pi}{4}$ , so the angle at the centre subtended by the same chord  $AB$  is

$$\angle APB = 2 \times \frac{\pi}{4} = \frac{\pi}{2}$$

So triangle  $APB$  is right-angled at  $P$ , with  $PA = PB = r$ . Also

$$|AB|^2 = (6-2)^2 + (5-1)^2 = 32$$

By Pythagoras,

$$r^2 + r^2 = 32$$

so  $r^2 = 16$ , giving

$$r = 4$$

(d) From part (b), the centre  $P$  lies on  $y = 7 - x$ , so write  $P = (a, 7 - a)$ .

From part (c),  $PA = 4$ , so

$$\begin{aligned}(a-2)^2 + (7-a-1)^2 &= 16 \\ a^2 - 4a + 4 + a^2 - 12a + 36 &= 16 \\ 2a^2 - 16a + 40 &= 16 \\ 2a^2 - 16a + 24 &= 0 \\ a^2 - 8a + 12 &= 0 \\ (a-2)(a-6) &= 0\end{aligned}$$

Hence the two possible centres are  $(2, 5)$  and  $(6, 1)$ .

To decide which is correct, test a point on one of the two circles. The point  $z = 10 + i$  lies on the circle centred at  $(6, 1)$ , diametrically opposite  $A$ :

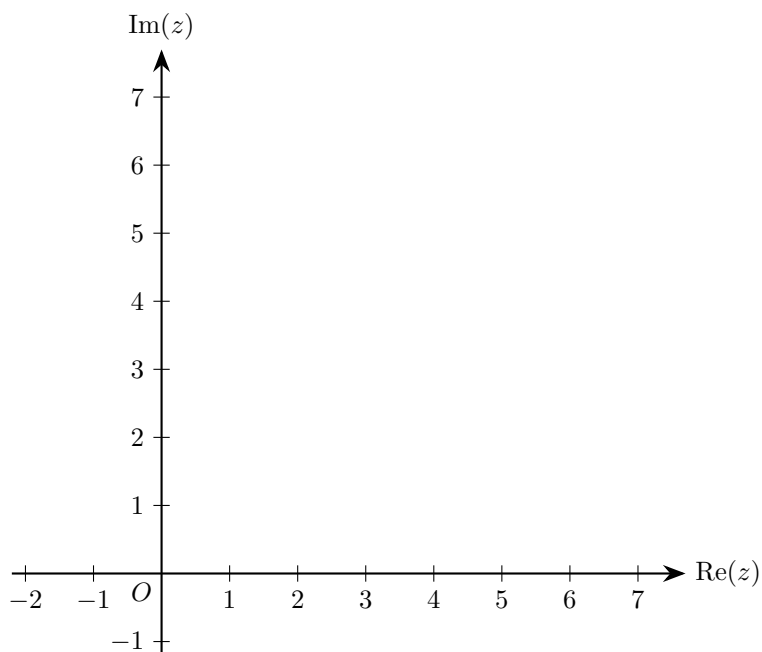
$$\begin{aligned}\arg \left( \frac{(10+i) - (2+i)}{(10+i) - (6+5i)} \right) &= \arg \left( \frac{8}{4-4i} \right) \\ &= \arg \left( \frac{8(4+4i)}{(4-4i)(4+4i)} \right) \\ &= \arg(1+i) \\ &= \frac{\pi}{4}\end{aligned}$$

So  $10 + i$  lies on the locus. But

$$\text{distance from } (10, 1) \text{ to } (2, 5) = \sqrt{(8)^2 + (-4)^2} = \sqrt{80} \neq 4$$

so it does not lie on the circle centred at  $(2, 5)$ . This also confirms that the arc lies below/right of  $AB$ .

Therefore the centre of  $C$  is  $(6, 1)$ .



10. (a) Sketch, on the Argand diagram below, the locus of points satisfying the equation

$$|z - (2 + 4i)| = 3 \quad [2]$$

- (b) Sketch, also on the Argand diagram, the locus of points satisfying the equation

$$\arg(z - 1) = \frac{\pi}{4} \quad [1]$$

- (c) For the complex number  $w$ , find the maximum value of  $|w|$  such that

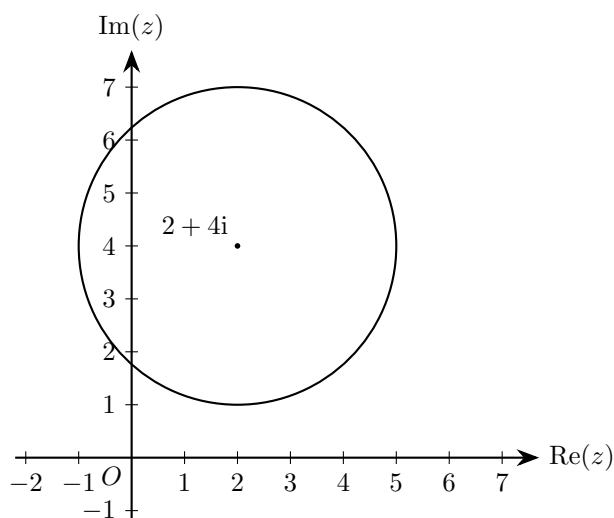
$$|w - (2 + 4i)| \leq 3 \quad \text{and} \quad 0 \leq \arg(w - 1) \leq \frac{\pi}{4} \quad [3]$$

### Solution

- (a) The locus

$$|z - (2 + 4i)| = 3$$

is a circle with centre at the point representing  $2 + 4i$  and radius 3.



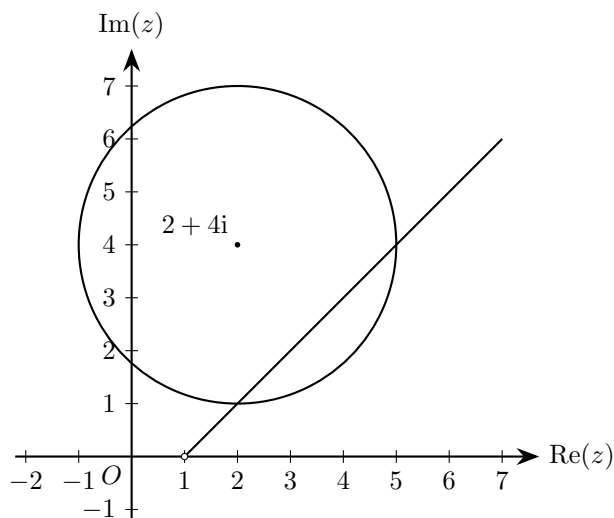
(b) The locus

$$\arg(z - 1) = \frac{\pi}{4}$$

is the half-line starting at the point representing 1 on the real axis, making an angle of  $\frac{\pi}{4}$  with the positive real axis.

In Cartesian form this is

$$y = x - 1, \quad x > 1$$



(c) Let

$$w = x + iy$$

The condition

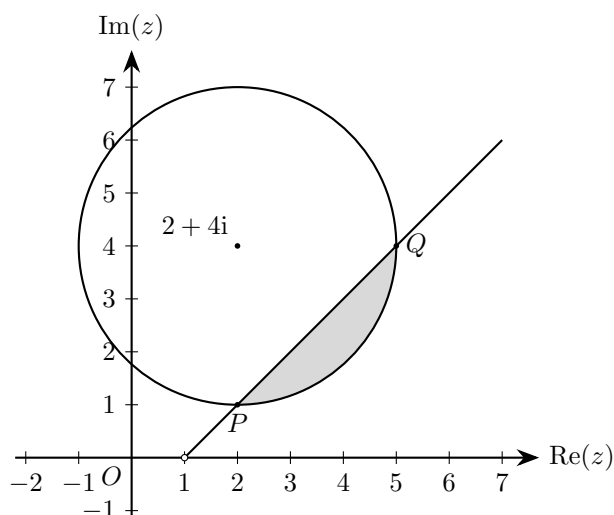
$$|w - (2 + 4i)| \leq 3$$

means that  $w$  lies inside or on the circle from part (a).

Also,

$$0 \leq \arg(w - 1) \leq \frac{\pi}{4}$$

means that  $w$  lies between the half-line from 1 along the positive real axis and the half-line from part (b).



Since the circle has lowest point  $(2, 1)$ , every point in the circle has  $y \geq 1$ , so the condition  $\arg(w-1) \geq 0$  is automatically satisfied.

Therefore the required region is the part of the disc lying on or below the line

$$y = x - 1$$

First find where this line meets the circle:

$$\begin{aligned}(x-2)^2 + (y-4)^2 &= 9 \\(x-2)^2 + (x-1-4)^2 &= 9 \\(x-2)^2 + (x-5)^2 &= 9 \\x^2 - 4x + 4 + x^2 - 10x + 25 &= 9 \\2x^2 - 14x + 20 &= 0 \\x^2 - 7x + 10 &= 0 \\(x-2)(x-5) &= 0\end{aligned}$$

so the intersection points are  $P(2, 1)$  and  $Q(5, 4)$ , as marked on the diagram.

Now  $|w|$  is the distance from  $O$  to the point representing  $w$ . Every point of the region lies in the disc, so  $x \leq 2 + 3 = 5$ . It also lies on or below the line, so  $y \leq x - 1 \leq 4$ . Both  $x$  and  $y$  are positive in the region, since  $y \geq 1$  and  $x \geq y + 1$ . Hence

$$|w|^2 = x^2 + y^2 \leq 5^2 + 4^2 = 41$$

Equality holds at  $Q$ , which is in the region.

Hence the maximum value of  $|w|$  is  $\sqrt{41}$ , attained when  $w = 5 + 4i$ .